Amidst the panic that ensued (backyard bomb shelters; atomic bomb drills for schoolchildren; McCarthy's "witch hunts" to root out spies, Communists, and their fellow travelers from government, army, media, and universities), a profound debate occurred amongst physicists and politicians. Edward Teller, one of the most innovative of the American atomic bomb design physicists, advocated a crash program to design and build the "superbomb" (or "hydrogen bomb")—a weapon based on the fusion of hydrogen nuclei to form helium. The hydrogen bomb, if it could be built, would be awesome. There seemed no limit to its power. Did one want a bomb ten times more powerful than Hiroshima? a hundred times more powerful? a thousand? a million? If the bomb could be made to work at all, it could be made as powerful as one wished.

John Wheeler backed Teller: A crash program for the "super" was essential to counter the Soviet threat, he believed. Robert Oppenheimer and his General Advisory Committee to the D.S. Atomic Energy Commission were opposed. It was not at all obvious whether a superbomb as then conceived could ever be made to work, Oppenheimer and his committee argued. Moreover, even if it did work, any super that was vastly more powerful than an ordinary atomic bomb would likely be too heavy for delivery by airplane or rocket. And then there were the moral issues, which Oppenheimer and his committee addressed as follows. "We base our recommendations [against a crash program] on our belief that the extreme dangers to mankind inherent in the proposal wholly outweigh any military advantage that could come from this development. Let it be clearly realized that this is a super weapon; it is in a totally different category from an atomic bomb. The reason for developing such super bombs would be to have the capacity to devastate a vast area with a single bomb. Its use would involve a decision to slaughter a vast number of civilians. We are alarmed as to the possible global effects of the radioactivity generated by the explosion of a few super bombs of conceivable magnitude. If super bombs will work at all, there is no inherent limit in the destructive power that may be attained with them. Therefore, a super bomb might become a weapon of genocide."

To Edward Teller and John Wheeler these arguments made no sense at all. The Russians surely would push forward with the hydrogen bomb; if America did not push forward as well, the free world could be put in enormous danger, they believed.